Standard deviation

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Example 8. *An incomplete frequency distribution is given as follows*

Variable	10-20	20-30	30-40	40-50	50-60	60-70	70-80	Total
No.of lab	12	30	◯	65	◯	25	18	229

$f_1 = 33.5$

$f_2 = 45.5$

Given that median value is 46, determine the missing frequencies using the median formula.

Example 9. *Median and Mode of the following wage distribution are known to be Rs 33.5 and Rs 32 respectively. Find the values of* f_3, f_4, f_5.

Wages	0-10	10-20	20-30	30-40	40-50	50-60	60-70	Total
frequency	4	16	f_3	f_4	f_5	6	4	230

Standard deviation: $x_1, x_2, x_3 \dots x_n$.

$E(X) = \bar{x}$

$E(X - \bar{x})$

(1) $$S.D = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$$

(2) A - assumed mean, $d = x - A$

$$S.D = \sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2}$$

(3) In case of discrete freq. dist.

$$S.D = \sqrt{\frac{\sum f_i x_i^2}{\sum f_i} - \left(\frac{\sum f_i x_i}{\sum f_i}\right)^2}$$

$\sqrt{\text{Variance}} = S.D.$

Ex: Consider the freq. dist.

x	4	10	12	13	17	20
f	2	8	9	4	3	2

Determine the variance & S.D.

$A = 12$, $d = x - A$

x	d	d^2	f	fd	fd^2
4	-8	64	2	-16	128
10	-2	4	8	-16	32
12	0	0	9	0	0
13	1	1	4	4	4
17	5	25	3	15	75
20	8	64	2	16	128

$$SD = \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2}$$

$d = \frac{x - A}{h}$, $$S.D = \sqrt{\left(\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2\right)} * h.$$

Ex: Find the SD

x	0–10	10–20	20–30	30–40	40–50
f	4	8	16	12	7

x	x_i	$x_i - A$ $= x_i - 25$	$d = \frac{x_i - A}{h}$	d^2	f	fd	fd^2
0–10	5	−20	−2	4	4	−8	16
10–20	15	−10	−1	1	8	−8	+8
20–30	25	0	0	0	16	0	0
30–40	35	10	1	1	12	12	12
40–50	45	20	2	4	7	14	28

$$S.D = h \times \sqrt{\frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f}\right)^2} = 11.47$$

Example 21. *Find SD for the continuous frequency distribution*

Wages	*10-20*	*20-30*	*30-40*	*40-50*	*50-60*
No.	*2*	*1*	*3*	*2*	*1*

Example 22. *For a group of 200 candidates, the mean and standard deviation of scores were found to be 40 and 15 respectively. Later on it was discovered that the scores 43 and 35 were missed as 34 and 53 respectively. Find corrected mean and standard deviation to the corrected figures*

Coeff of variation: $\frac{\sigma}{\bar{x}} \times 100$

Example 2. *A factory produces 2 types of electric bulbs A and B. In an experience relating to their life, the following results were obtained*

lengths in life	*500-700*	*700-900*	*900-1100*	*1100-1300*	*1300-1500*
No. of bulbs in A	*5*	*11*	*26*	*10*	*8*
No. of bulbs in B	*4*	*30*	*12*	*8*	*6.*

Soln:

$\bar{x}_A =$ $\qquad$ $\bar{x}_B =$

$\sigma_A =$ $\qquad$ $\sigma_B =$

Assume $A = 1000$

$h = 200$

$$\bar{x}_A = A + \frac{\Sigma fd_i}{\Sigma f} \times h$$

$$\sigma_A = \sqrt{\frac{\Sigma fd_i^2}{\Sigma f_i} - \left(\frac{\Sigma fd_i}{\Sigma f_i}\right)^2} \times h$$

x_i	f_i	$d_i = x_i - A$	$d_i = \frac{x_i - A}{h}$	fd_i	fd_i^2
600	5	−400	−2	−10	20
800	11	−200	−1	−11	11
1000	26	0	0	0	0
1200	10	200	1	10	10
1400	8	400	2	16	32

1000	20	0	0	0	0
1200	10	200	1	10	10
1400	8	400	2	16	32

$\bar{x}_A = 1016.6$, $\sigma_A = 219.9$

$$C.V_A = \frac{\sigma}{\bar{x}_A} \times 100 = 21.6$$

Assume $B = 800$

$h = 200$

x_i	f_i	$d_i = x_i - B$	$d_i = \frac{x_i - B}{h}$	fd_i	fd_i^2
600	4	-200	-1	-4	4
800	30	0	0	0	0
1000	12	200	1	12	12
1200	8	400	2	16	32
1400	6	600	3	18	52

$\bar{x}_B = 860$, $\sigma_B = 220$, $CV_B = 26.3$.

$CV_A \lessgtr CV_B$.

$$E(X) = \frac{\sum f_i x_i}{\sum f_i}$$

$$E(x-\bar{x})^n = \frac{\sum f_i (x_i - \bar{x})^n}{\sum f_i}$$

Central moment: $\mu_n = E[X - \bar{X}]^n$

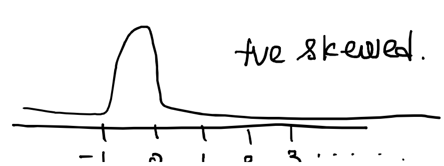
$$\mu_1 = E(x - \bar{x})$$
$$= E(x) - E(\bar{x})$$
$$= \cancel{E(x)} - \cancel{\bar{x}} = 0$$

$$\mu_2 = E(x-\bar{x})^2 = E[x^2 - 2x\bar{x} + \bar{x}^2]$$
$$= E(x^2) - 2E(x\bar{x}) + E(\bar{x}^2)$$
$$= E(x^2) - 2\bar{x}\,E(x) + \bar{x}^2$$
$$= E(x^2) - 2E(x)^2 + E(x)^2$$
$$= E(x^2) - [E(x)]^2 = \text{Var}\,X.$$

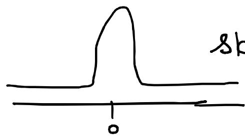
Applications related to moments.

x	-1	0	1	2	3 ....

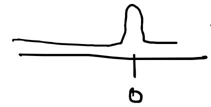
+ve skewed.



x -10 -9 . . . . -1 0 1



skewed.
or symmetric

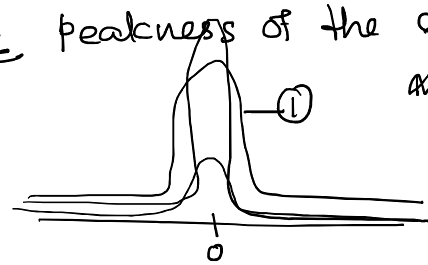


-vely skewed.

skewness : $\dfrac{\mu_3^2}{\mu_2^3}$
β

1. $\beta > 0 \Rightarrow$ the curve is +vely skewed
2. $\beta < 0 \rightarrow$ the curve is -vely skewed
3. $\beta = 0 \Rightarrow$ the curve is sym.

kurtosis: peakness of the distrubtion



$\alpha = \dfrac{\mu_4}{\mu_2^2}$

1. $\alpha = 3$, the curve is mesokurtic.
2. $\alpha < 3$, the curve is platykurtic
3. $\alpha > 3$, the curve is leptokurtic.

Eg: Let First four moments of the distribution are 0, 0.6, 0.8, -0.9.

Determine skewness & kurtosis.

$\mu_1 = 0$, $\mu_2 = 0.6$, $\mu_3 = 0.8$, $\mu_4 = -0.9$.

$\beta = \dfrac{\mu_3^2}{\mu_2^3} > 0 \Rightarrow$ the dist is +ve skewed

$\alpha = \dfrac{\mu_4}{\mu_2^2} = \dfrac{-0.9}{0.6^2} < 3$, $\Rightarrow$ the curve is platykurtic.